

The SteamHead Design Immersion Curriculum

**A Collaborative Project Supporting Maker Education
and Design Thinking in the Classroom**

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Made Possible By:

[SteamHead](#), [MakerSAIS](#), [MakeFashion Edu](#) and [Cultivative](#)



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Introduction

The SteamHead Design Immersion Curriculum builds upon the established educational values of existing design and technology curriculums to instill creativity, critical thinking, and problem solving skills to students across all subjects in a holistic program to prepare them for the challenges of the future. The curriculum is meant to teach students all the tools they need to be innovative creators: from the high-level concepts of design planning and community impact all the way through to the hands-on digital and physical tools needed to enact their ideas. Our lessons are defined and measured in the learning pathways found in this document via a series of “badges” that represent the level and degree of accomplishment in a given skill as the students progress. This method allows the most freedom and flexibility to students as possible.

‘Immersion’ best encapsulates our integrated approach towards technological and design education. It’s meant to reflect some of the value and efficacy of language immersion education because, like language, design thinking and technology are embedded in nearly every aspect of educational, personal, and business life. To that end, design immersion is meant to free these concepts from the barriers of traditionally siloed educational subjects and to cultivate creative problem solving as a dialect of learning.

Our curriculum has developed and changed over several years as we implemented and experimented in the classrooms at Shenzhen American International School (SAIS) with the support of the amazing teachers there. SAIS has made Project Based Learning (PBL) central to their curriculum and philosophy and became the first entirely PBL school in Shenzhen, so it was the ideal environment to develop our curriculum. We expect to continue to innovate and develop it both at SAIS and in our own classes with the help of the larger education community.

The SteamHead Design Immersion Curriculum is a collaboration between SteamHead, MakerSAIS, MakeFashion Edu, and Cultivative. We present the curriculum here as a living document, so that others in the education community might benefit from our efforts. If you would like to use this information to help your own communities, please do so under Creative Commons Noncommercial ShareAlike license, which can be found here:

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Learning Pathways

Design thinking requires both a thorough understanding of abstract heuristics as well as hands-on interactive skills. We are building on the Stanford d.school's Design Thinking Process of "empathize, define, ideate, prototype, and test," as well as a version of the "head, heart, hands" model, to ensure that students grasp the individual concepts of design process, how these skills can be integrated into any project or problem, and how to approach design with empathy. To that end, our four learning pathways, Community Impact, Design Planning, Digital Tools, and Physical Tools, divide skills into sensible categories for definition but should also be understood as one, gestalt design and creation process. E.g., a student's hands-on experience with physical tools should inform their design process when considering possible materials and limitations for a prototype, and vice-versa with design process informing what skills might be needed for a project.

Community Impact

Community Impact is the most important, but also the most esoteric, pathway in our curriculum because it addresses the need for meaning and purpose in students' work. One of the failings of older education models was their reliance on rote problem solving and artificial practice problems. Students can learn academic topics this way, but the ideas will feel disparate, not applicable to the real world, and, most of all, not interesting or important to the students and their community. Design thinking needs to begin with empathy not only as a virtue, but also as a pathway to listening and thinking about the needs of those around us then defining a problem to solve. Students share their work as the final aspect of this pathway to encourage self reflection and to reinforce that their project has impact outside the classroom. Starting from the perspective of community impact is the pathway that leads students to define meaningful projects, be motivated to innovate, and instill meaning into the work they do at school and beyond.

Design Process

The Design Process pathway represents the methods students need to ideate, manage, and revise their projects. This is the second of our abstract pathways, but it should be accessible and familiar to anyone who has tried to complete a task with more than one step: the need for planning and process becomes more and more apparent with each layer of complexity. The Design Process pathway allows students to pick up the problem they defined in Community Impact and attempt to solve it. They first ideate possible solutions or avenues of exploration and make drafts, then they organize the steps, tools, and other factors needed to create a prototype. Through practicing and organizing their own projects, students gain independence in setting up and cleaning up their work as well as become empowered to solve problems they choose rather than waiting to be told what to work on. This pathway will also teach students the meta-structure of design thinking and the purpose of the abstract scaffolding they are being taught both as constructs and as applied skills. For example, students would write a business plan for their prototype product that requires an up front understanding of the project's resources, scope, and scale. Last, this pathway teaches iteration and testing: the important process of building on and learning from one's work; the essential lesson that, contrary to the traditional mindset of producing "right or wrong answers," we often learn most from the trials and prototypes that don't work, rather than ones that do.

Fabrication Tools

Fabrication tools are often one of the first things that come to mind when the topic of project based learning and maker education come up, and understandably so, because few things inspire and excite us more than seeing an idea come to life in physical form whether it's on the tray of a 3D printer or the woodshop table. While that base-level excitement fuels the creative and learning process, the Fabrication Tools pathway also serves as a grounding point and a means to solve problems identified in the Community Impact pathway and require forethought, attention to detail, safety, and management of resources. The classroom provides an environment where students can learn these tools under guidance

and supervision, so they have the skills and respect for fabrication tools to use them in future projects and for the rest of their lives. Regardless of whether students learn tools and skills held by humans for hundreds of years like papercraft, woodcraft, and stitching, or they learn the kind of technological tools hyped in the headlines like 3D printers, laser cutters, and block circuits: few lessons are more powerful than when students become empowered to literally change the world around them.

Digital Tools

Digital Tools have transformed our world and made this the Digital Age, and yet individual apps, websites, and coding languages are quickly swallowed up in the shifting sands of popularity and technological advancement, so this pathway has to focus on the kind of digital literacy, frameworks, and heuristics needed to use any kind of digital tool in addition to those in front of us today. Our students often arrive to the classroom having been steeped in the digital world from birth, and the trope of adults needing to “ask a young person” how their own digital tools work shouldn’t surprise us. However, it’s naive and dangerous to assume that students don’t need a thorough education on this topic. The hazards of data security and privacy jump first to mind, but even common skills like file management can seem outmoded and unnecessary until they become a crisis of lost projects. Like the native language skills students bring to the classroom, students can already ‘speak’ with digital tools, but they need to be surrounded by educators modeling positive habits in order to develop that skill. The notion that classrooms should remove digital tools for being ‘distractions’ or to better serve traditional pedagogy amounts to hiding our heads in the sand in the face of an already digital world. Pedagogy, teachers, and students all need to adapt to these changes because digital tools open up entirely new modes of thinking and working, allow access to near limitless information, and are 21st century implements of innovation; we must empower our students with these tools if they are to be ready for the future.

System of Badges

Our Design Immersion Curriculum relies on a system of digital badges to measure student progress, allow for open learning pathways, and encourage self documentation. The badges can be implemented on a number of open platforms such as Badg.us, Badgr, and Mozilla's Open Badge Infrastructure (OBI). These are systems for sharing a unique digital artifact, creating something similar to a scout badge, that consists of an image and metadata about the criteria, issuer, and evidence of the work done to earn it. The badge is then added to a student's portfolio and can be brought to future classes or shown as a history of accomplishments. Research by Mozilla and ISTE among others has shown these badges to be an effective, evidence-based system to measure student performance based on actions taken, not facts memorized, which is important for the subjective topic of design. Digital badges encourage student agency as they strive toward accomplishable goals that excite them. Once achieved, the student engages in self documentation and reflection, which is meaningful for the learning process and necessary to reduce the resource cost for the teacher tracking multiple students on multiple projects. An introductory level implementation might involve attaching a badge to a lesson that all students participate in, but we plan for Design Immersion to have open-ended objectives that allow students to engage in true design thinking and explore new pathways. As they progress, students self document and 'apply' for badges, so teachers can measure and reward progress in skills even while efforts toward a finished product fail or are abandoned. It's far too easy to see design projects (and other open projects such as in art or engineering) only through the lens of the end result, but there is true, measurable, educational value in the exploration and drafting phases of design that can be captured by our badging system.



Work in Progress Content

We have decided to keep our WIP Content included in the Wiki document, so you can see what we are developing!

Smart Homes Project Example

Ms. Leung, teaching Grade 3, talks with her class about projects that could be done during the year. Grade 3 is kind of young, so she comes up with several ideas based on student interest,

Class comes up with the idea together, and concur on Smart Homes. Ms. Leung builds the project and scaffolds in the academics

But the class comes to a consensus on which project to choose. They decide on “Smart Homes” is going to use a 2 month long project called “Smart Homes.” The class talks about the end product (what is the problem, who are the people you are helping), together this is decided, with guidance from the teacher as a project to aid her students’ learning. Smart Homes will involve students studying the environment and people inside their homes, and developing an electronic solution to a problem that the students identify.

This project will involve the Design Immersion curriculum, but it will also involve academic subjects - Humanities, Math, English. The Design Immersion goals will overlay some of the academic lessons during the Smart Homes project.

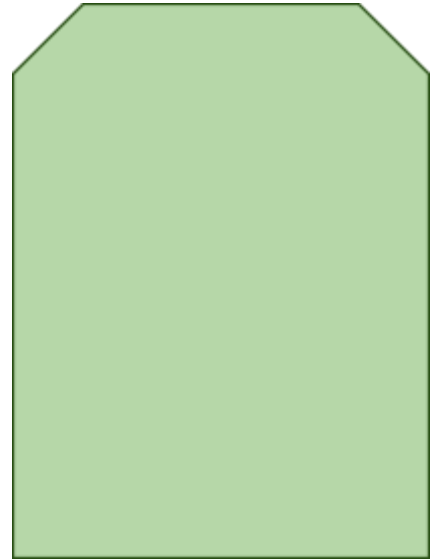
The class doesn’t move together, each student works on their own project at their own pace.

For example, students will interview their family (a badge and additionally a language topic) to better understand them during the development of a Smart Homes product, and will research the history (a badge and additionally a humanities topic) of their family members as a followup to the interviews.

Ms. Leung can either tell students to reference the online badge system, or print out paper activities for each student. She chooses to print, as she will be asking students to document some steps on each activities prior to creating their digital submission. To print the activities, Ms. Leung opens the prepared activities her school shares, but types some customized instructions onto it to better suit her students and goals.

Ms. Leung plans to use 8 badges, and for the project to take 40 class hours over 2 months (5 hours per week). During the project, students will use more skills that are contained in the 8 badge worksheets, however each badge requires teacher review, and 8 badges is an amount of review that Ms. Leung is comfortable with.

During the projects launch, activities are available to the class. Ms. Leung additionally gives her students the option to visit the Badge Worksheet binder, at any point during the first month of the project. This binder is full of various Badge Worksheet printouts, and students are allowed to exchange their badges within the same Pathways. For example exchanging an Interview badge for a User Stories badge, both of which help the student better understand their subjects and problems to solve.



Here are the 8 badges Ms. Leung as initially distributed

- Community Impact: Interview, Roleplay
- Design Planning: Drafting, Business Plan
- Fabrication Tools: Block Circuits, Papercraft
- Digital Tools: Videography, Hacking

She has, as an optional step, customized several of the badge documents

- Community Impact: **iPad Video Interview**, Roleplay
- Design Planning: Drafting, Business Plan
- Fabrication Tools: **M5 Stack IoT Circuits, Cardboard Papercraft**
- Digital Tools: **iMovie Videography**, Hacking

After the project introduction, Ms. Leung's class transition into their regular subjects, which will be integrated with the Design Immersion project. During English class they have a writing assignment, which is geared towards preparing for their Interview badge. During Math class, they use graph paper to draw and label measurements, which will contribute towards their Drafting badge.

During each day's subject, the teacher knows the badges and the project that the students are pursuing, and orients her lessons to support that project. Students are explicitly informed of this connection during the lessons, and are given opportunities to document their work under their project's folder.

FAQ

- What are Badges?
 - a. Badges are evidence based systems measure student performance based on actions taken, not facts memorized, which is important for the subjective topic of design
 - b. Badges are co-owned co-operative artifacts, allowing for an increase of student agency in learning objectives
- In what ways are Badges best for a Design curriculum?
 - a. With design, it's very tempting to measure only the results of a project, but the main learning outcomes are achieved during the building of the project. Badges help teachers measure the effort put into a project, not only the end result.
 - Badges allow for plentiful self-documentation, which is necessary if numerous steps of design are to be assessed, because otherwise the resource cost is too high for the teacher to assess all the steps of all the student.
 - Traditionally this is overcome by having students all follow the same pathways in a lesson, however in design this must necessarily not-be the procedure because we are also assessing the student's ability to create new pathways.
 - Student self documentation must be summarized and classified within a system, so that teachers can assess students systematically.
- Can you give me a Twitter-sized explanation?
 - a. The Media Research badge, in the Empathy pathway, has students document what research was done, capturing how the media was accessed, searched, and what was discovered.

b. The Hacking badge, in the Digital Tools pathway, has students download a resource or file that fits the teacher's requirements, but the student has a very wide berth to discover new resources that can be hacked. Massive self agency on discovering and improving/hacking that resource, plus applying it to the problem at hand.

- Explain the 4 learning pathways chart
- [\[notes moved to other doc\]](#)
- [http://www.susted.com/wordpress/content/head-heart-and-hands-model-for-transformative-learning-place-as-cont
ext-for-changing-sustainability-values_2015_03/](http://www.susted.com/wordpress/content/head-heart-and-hands-model-for-transformative-learning-place-as-context-for-changing-sustainability-values_2015_03/)
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